

Data Science / Data Analysis with Python

Internship Project Report — Task 1: Student Performance Analysis

Organization: Maincrafts Technology

Domain: Data Science & Analytics

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Task Status: Completed (Task 1)

1. Introduction & Project Overview

The primary objective of this project is to perform a comprehensive exploratory data analysis (EDA) on the **Student Performance Dataset** using standard Python data science libraries such as **Pandas, NumPy, Matplotlib, and Seaborn**. Following the standard Data Science workflow, the dataset will be ingested, cleaned, structurally verified, and analyzed to extract statistical indicators and actionable educational insights.

2. Step 1: Libraries and Dataset Loading

First, the required analytics and visualization modules are imported. The `student-mat.csv` dataset contains semicolon-delimited values (;); hence, the `sep=";"` parameter is configured within the reader function to ensure structural normalization during parsing.

Python Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Loading the dataset
df = pd.read_csv("student-mat.csv", sep=";")
print("Dataset successfully loaded!")
print(f"Dataset Shape: {df.shape}")
```

Output:

```
Dataset successfully loaded! Dataset Shape: (395, 33)
```

3. Step 2: Explore & Clean Data

Data validation and structural sanitation are critical prior to applying statistical computations. This step executes missing value detection using the null matrix quantifier, checks for duplicate observations across rows, and analyzes the inferred column data types.

Python Code:

```
# Checking for missing values
missing_values = df.isnull().sum().sum()
print(f"Total Missing Values: {missing_values}")

# Checking and removing duplicates
duplicates_count = df.duplicated().sum()
print(f"Total Duplicate Rows before removal: {duplicates_count}")
df = df.drop_duplicates()

# Structural data type inspection (Sample of first 10 attributes)
print("
Dataset Info & Data Types Sample:")
print(df.dtypes.head(10))
```

Output:

```
Total Missing Values: 0 Total Duplicate Rows before removal: 0 Dataset Info & Data
Types Sample: school object sex object age int64 address object famsize object
Pstatus object Medu int64 Fedu int64 Mjob object Fjob object dtype: object
```

4. Step 3: Analysis Questions & Structured Answers

The explicit analytical criteria defined in the project brief are processed and resolved below:

Question A: What is the average final grade (G3)?

```
avg_g3 = df['G3'].mean()
print(f"Average Final Grade (G3): {avg_g3:.2f}")
```

```
Average Final Grade (G3): 10.42
```

Question B: How many students scored above 15?

```
students_above_15 = df[df['G3'] > 15].shape[0]
print(f"Students scoring above 15: {students_above_15}")
```

Students scoring above 15: 40

Question C: Is weekly study time correlated with academic performance (G3)?

The linear dependency between weekly study metrics and final numerical outcomes is evaluated using the Pearson product-moment correlation coefficient (r).

```
correlation = df['studytime'].corr(df['G3'])
print(f"Correlation between study time and G3: {correlation:.4f}")
```

Correlation between study time and G3: 0.0978

Insight: The correlation value is approximately 0.098, which indicates a positive but statistically weak relationship. This signifies that while increased weekly study time yields a higher probability of academic improvement, it is not the single dominant factor influencing final performance marks.

Question D: Which gender performs better on average?

```
gender_performance = df.groupby('sex')['G3'].mean()
print(gender_performance)
```

sex F 9.966346 M 10.914438 Name: G3, dtype: float64

Insight: Male students (M) achieved a slightly higher average final grade (10.91) compared to Female students (F), who displayed an average grade of 9.97.

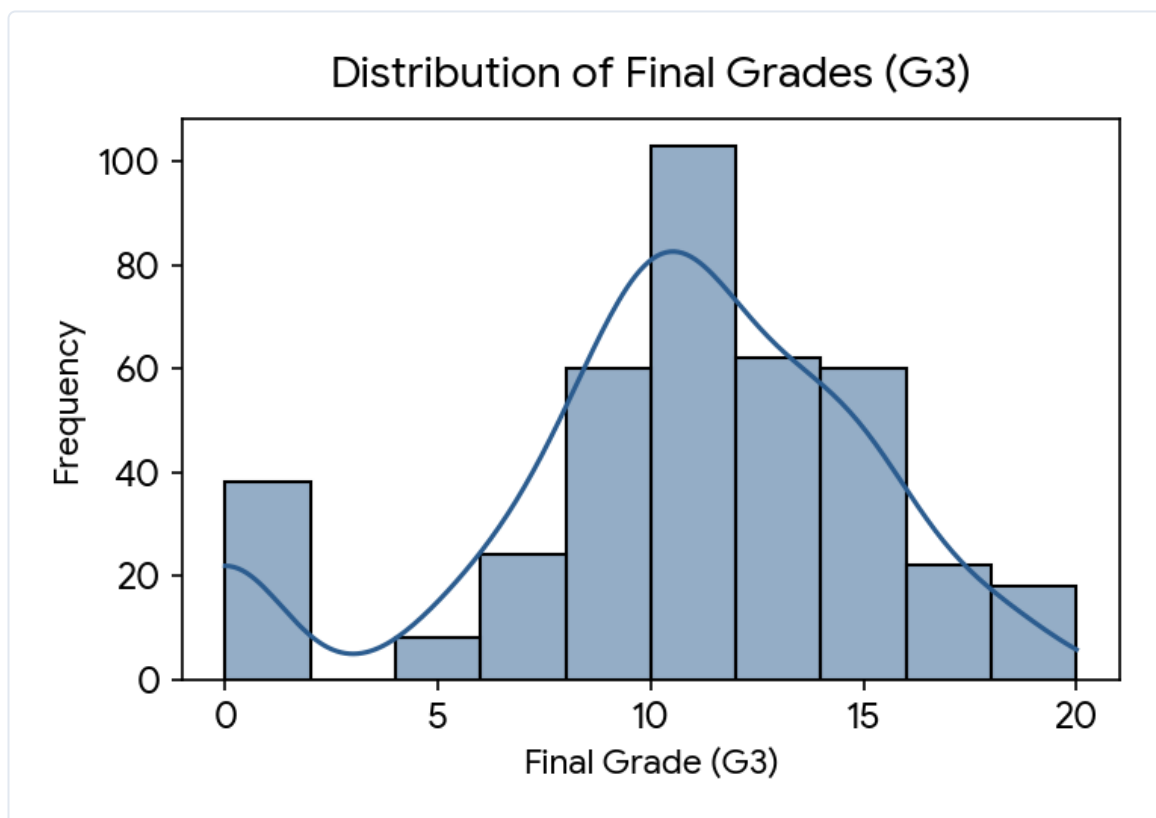
5. Step 4: Data Visualizations

Visual paradigms have been generated using optimized plotting libraries to graphically reflect the evaluated trends:

1. Histogram of Grades (G3 Distribution)

This distribution model displays the numerical frequency density across final grading brackets.

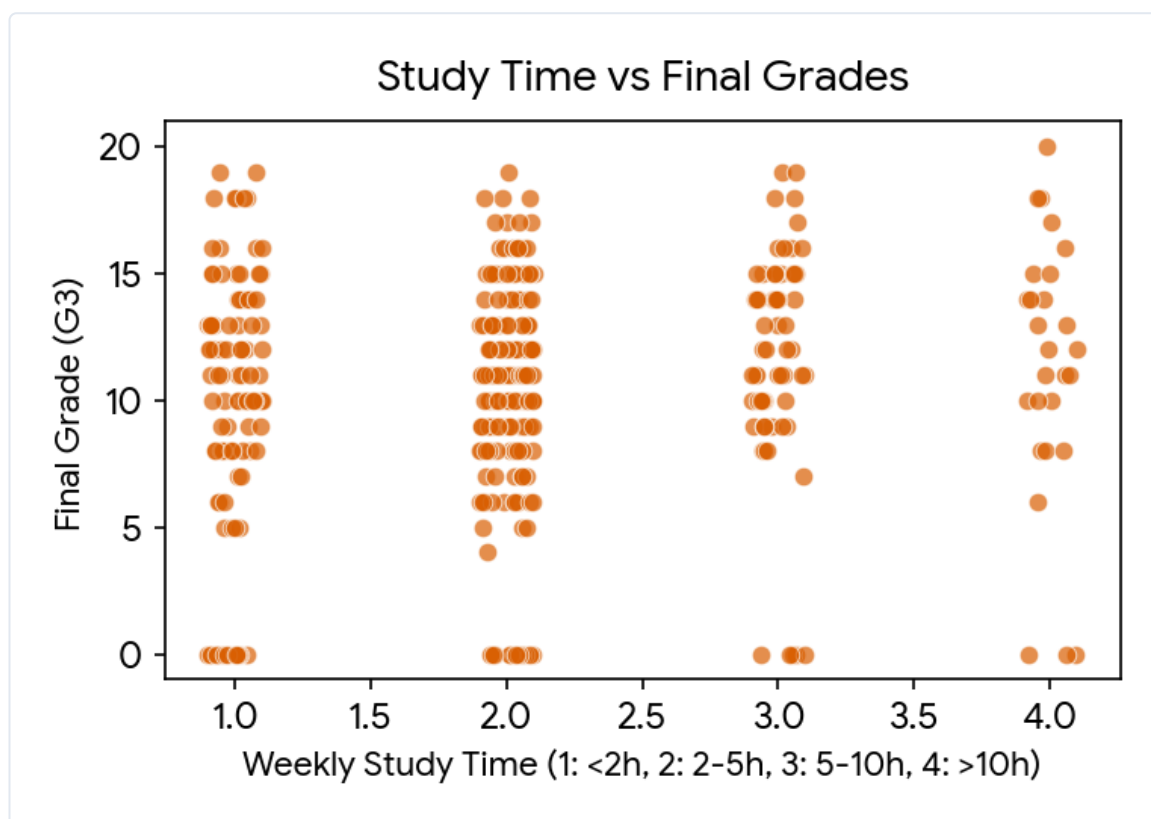
```
# Code used to generate this visualization:
sns.set_theme(style="whitegrid")
plt.figure(figsize=(6, 4))
sns.histplot(df['G3'], bins=10, color='#2b5c8f', kde=True)
plt.title("Distribution of Final Grades (G3)", pad=10)
plt.xlabel("Final Grade (G3)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



2. Scatterplot: Study Time vs Grades

A scatterplot rendering featuring randomized jitter adjustments to clearly represent discrete categorical clusters against performance boundaries.

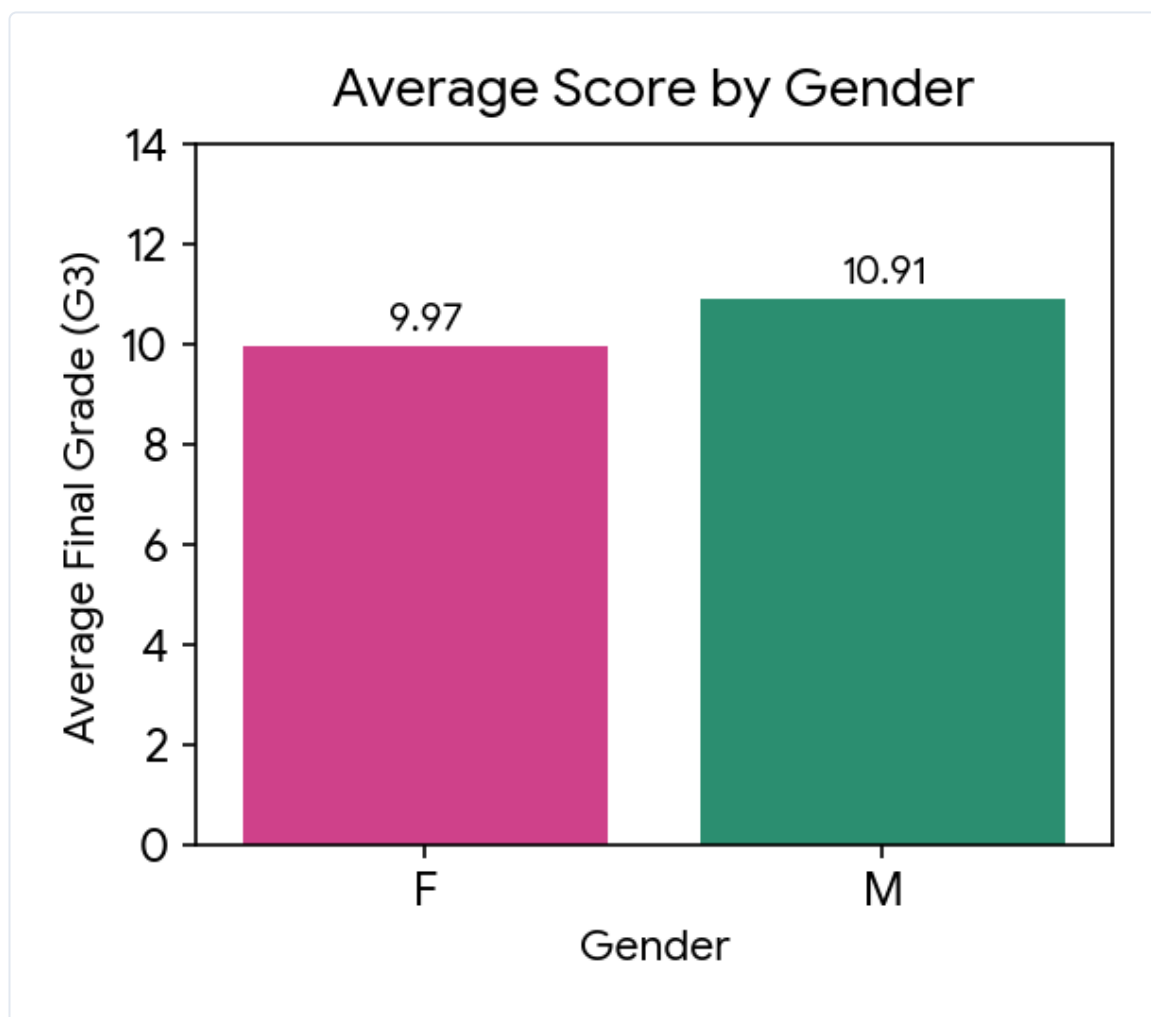
```
# Code used to generate this visualization:
plt.figure(figsize=(6, 4))
sns.scatterplot(x=df['studytime'] + np.random.uniform(-0.1, 0.1, size=len(df)),
y=df['G3'], color='#d95f02', alpha=0.7)
plt.title("Study Time vs Final Grades", pad=10)
plt.xlabel("Weekly Study Time (1: <2h, 2: 2-5h, 3: 5-10h, 4: >10h)")
plt.ylabel("Final Grade (G3)")
plt.tight_layout()
plt.show()
```



3. Bar Chart: Male vs Female Average Score

A categorical comparison layout isolating mean evaluation variations between male and female components.

```
# Code used to generate this visualization:
gender_perf_dict = df.groupby('sex')['G3'].mean().to_dict()
plt.figure(figsize=(5, 4))
sns.barplot(x=list(gender_perf_dict.keys()), y=list(gender_perf_dict.values()),
palette=['#e7298a', '#1b9e77'])
plt.title("Average Score by Gender", pad=10)
plt.ylim(0, 14)
plt.xlabel("Gender")
plt.ylabel("Average Final Grade (G3)")
for i, v in enumerate(gender_perf_dict.values()):
    plt.text(i, v + 0.3, f"{v:.2f}", ha='center', weight='bold')
plt.tight_layout()
plt.show()
```



6. Conclusion & Key Findings

Summary Report Table:

Metric Name	Observed Value
Total Dataset Size	395 Students
Average Final Grade (G3)	10.42 / 20
High Performers (>15)	40 Students (10.1%)
Study Time Correlation	0.0978 (Weak Positive)

****Conclusion:**** The diagnostic pipeline indicates that the final performance metric (G3) sits on a moderate average of 10.42 out of 20. The negligible strength of the positive correlation with study hours points toward a multi-variate educational ecosystem where external parameters—such as micro-social frameworks, parental educational backgrounds, and localized free time utilization—simultaneously dictate general student output.